ArcCanvas

Curved VR UI Toolkit — Documentation

Bend any Unity Canvas into a curved, fully interactive surface. Original mesh-warping and interaction system — no third-party dependencies.

Version 1.0

Unity 2022.3 LTS and newer · Built-in / URP / HDRP

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1. Overview

ArcCanvas bends any Unity Canvas into a curved surface — Cylinder, Ring, Sphere, Wave, Cone, Spiral, or Ellipse — while keeping every UI element fully interactive. Unlike shader-only bending effects, ArcCanvas warps the real UI mesh and generates a matching invisible interaction surface, so clicks, gaze, and VR controller rays land exactly where they visually appear.

Key capabilities:

- 7 curve shapes with concave/convex direction control
- Full mouse, gaze, and VR controller support (Meta XR, OpenXR, XR Interaction Toolkit)
- TextMeshPro support, including nested inside Buttons
- One-slider Simple Mode plus full Advanced Mode control
- Built-in performance system: distance-based LOD and an optional GPU bend shader
- Animated transitions and one-click presets
- Zero manual wiring — every child element is detected and bent automatically

2. Installation

Import the ArcCanvas folder into your project's Assets directory. No additional setup or dependencies are required.

If TextMeshPro is installed in your project (Window ■ TextMeshPro ■ Import TMP Essential Resources), ArcCanvas automatically detects it and enables TMP-specific bending support — no configuration needed.

3. Quick Start

1. Add **ArcCanvasSettings** to a Canvas. It automatically switches the Canvas to World Space render mode, which is required for the curve to be visible.
2. Add **ArcCanvasInteractionRaycaster** to the same Canvas (replace the stock GraphicRaycaster if present) and assign a camera or VR pointer.
3. That's it. Every child Graphic — Image, Button, Text, TextMeshPro, RawImage — is automatically bent and automatically clickable.
4. Drag the Curve Amount slider in the Inspector to see it bend live, in both Edit Mode and Play Mode.

4. Simple Mode vs. Advanced Mode

ArcCanvas defaults to **Simple Mode**: a single **Curve Amount** slider (0 = flat, 1 = fully curved) drives everything, mapped internally to whichever parameter actually controls curvature for the selected shape.

Shape	What Curve Amount Controls
Cylinder / Ring / Sphere / Cone / Spiral	Radius (smaller radius = tighter curve)
Wave	Wave amplitude
Ellipse	Both X and Z radii, proportionally

Switching to **Advanced Mode** (Inspector checkbox, or the runtime panel's "Show Advanced Controls" button) hands over direct control of every parameter: exact radius, curve direction, per-shape fields, subdivision quality, and the optional depth multiplier.

5. Component Reference

Component	Purpose
ArcCanvasSettings	Main component. Add to a Canvas to bend it. Controls shape, curve amount, quality, and blend.
ArcCanvasInteractionRaycaster	Replaces GraphicRaycaster. Generates a matching invisible collider so clicks/gaze/VR rays land correctly on the curved surface.
ArcCanvasVertexEffect	Auto-added to each Graphic. Subdivides and warps its mesh. Not added manually.
ArcCanvasTMPBendAdapter	Auto-added to TextMeshPro components when TMP is installed. Warps TMP's per-character vertex data directly.
ArcCanvasLOD	Optional. Reduces subdivision quality by camera distance and skips rebuilds when off-screen.
ArcCanvasTransitionAnimator	Optional. Animates curve blend, radius, or morphs between shapes at runtime.
ArcCanvasRuntimeController	Optional. Auto-builds an in-game control panel (Simple + Advanced views) for live tuning.
ArcCanvasHaptics	Optional. Sends controller haptic pulses on hover/click for VR.

6. Shape Parameters Reference

Field	Applies To	Description
radius	Cylinder, Ring, Sphere, Cone, Spiral	Controls how tightly the canvas wraps. Smaller = tighter curve.
curveDirection	All shapes	Away (concave, wraps around viewer) or Toward (convex, bulges toward viewer).
waveAmplitude / waveFrequency	Wave	Height and repeat rate of the ripple.
coneTaper	Cone	0 = cylinder, 1 = fully tapers to a point.
spiralTurns / spiralHeightPerTurn	Spiral	Number of wraps and vertical rise per wrap.
ellipseRadiusX / ellipseRadiusZ	Ellipse	Independent horizontal and depth radii.
useCustomDepth / depthMultiplier	All shapes (optional)	Off by default. Scales curve depth independently of radius.

Every shape function returns the exact canvas center as its unmoved anchor point — the middle of the canvas never shifts; only the edges bend away from or toward it.

7. One-Click Presets

Preset	Shape Used	Effect
Subtle Curve	Cylinder	Large radius (1600). Gentle, barely-there curve.
Wrap Around Viewer	Cylinder	Tight radius (600). Strong immersive wraparound.
Curved Monitor	Cylinder (convex)	Radius 1000, bulges toward the viewer like a curved display.
Space Helmet	Sphere	Radius 900. Dome-like HUD feel.
Flatten	—	Sets curve amount / blend to 0.

Presets switch off Simple Mode automatically, so their exact values are not immediately overwritten by the Curve Amount slider mapping.

8. Performance

Tool	When To Use
ArcCanvasLOD	Multiple curved panels in one scene. Lowers subdivision quality by distance and skips rebuilds off-screen.
ArcCanvasGPUBend.shader	Panels that update every frame (scrolling lists, live counters). Moves bending into the vertex shader.
Quality field	Lower for simple flat-colored panels; raise only where smoothness is visibly needed.

9. TextMeshPro Support

TextMeshProUGUI generates and pushes its own mesh directly, bypassing Unity UI's standard mesh-modifier pipeline that Image and legacy Text use. ArcCanvas detects TMP components automatically and attaches ArcCanvasTMPBendAdapter instead, which warps TMP's per-character vertex data using TMP's own supported API. This works correctly even when TMP text is nested inside a bent Button.

No setup is required — this only activates when the TextMeshPro package is present in your project.

10. Troubleshooting

Symptom	Cause	Fix
Nothing bends at all	Canvas is not World Space	ArcCanvasSettings auto-fixes this by default. Confirm Auto Force World Space is enabled.
TMP text doesn't bend, but Images/Buttons do	TMP bypasses the standard mesh pipeline	Handled automatically by ArcCanvasTMPBendAdapter — ensure TMP Essentials are imported.
Changes don't update live	Component was recently added and hasn't refreshed	Click Force Rebuild Now in the Inspector, or call RefreshChildEffects().
Clicks land in the wrong place	Interaction raycaster missing or misconfigured	Add ArcCanvasInteractionRaycaster and assign a camera or VR pointer.
Console warning about Quality being reduced	Element has many triangles at high Quality	Expected safety behavior — lower the global Quality slider for a uniform look, or ignore it.